

Phu Vinh Nguyen

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SKILLS

Orchestration: Airflow 3, Kestra, Docker, Terraform, Kafka, Git, HPC environments, LIMS, Docker, Git

Programming Languages: Python, JavaScript, SQL, HTML, CSS, Bash, R

Application Development: ReactJS, NodeJS, ExpressJS, ElectronJS, CustomTkinter, Tesseract

Data Processing & Storage: PySpark, BigQuery, GCS, AWS S3, PostgreSQL, MySQL

Analytics: Jira, Power BI, Google Looker Studio

AI Tools: AI Prompting, Gemini, Ollama, Tesseract

WORK EXPERIENCE

Lab Automation Engineer (Contractor) | Sanofi Pasteur

Jun 2025 – Present

- Developing an internal ticketing application using ElectronJS, ReactJS, and NodeJS to replace ad-hoc verbal R&D requests with a structured digital workflow, routing problem reports to the automation engineering team via SMTP for traceable and prioritizable resolution.
- Actively building a ticket monitoring dashboard using WebSocket to track real-time ticket status and resolution process for automation teams.
- Modernized lab automation systems by implementing Git version control workflows and safeguarding pipeline integrity across experimental workflows

Automation Engineer | Foundation Medicine Inc

Nov 2024 – Apr 2025

- Reduced support ticket volume by 20% by delivering a self-service Python GUI using CustomTkinter that empowered R&D staff to independently manage instrument naming configurations without engineering intervention.
- Engineered a ReactJS visualization tool that parsed and rendered Hamilton robot configuration metadata, giving teams immediate insight into system parameters
- Designed and delivered ETL pipelines to onboard sample data into Sapio LIMS, establishing automated tracking from intake to processing and eliminating a major source of manual entry errors in R&D workflows.

Lab Automation Engineer | Sanofi Pasteur

Oct 2022 – Nov 2024

- Drove a 30% reduction in assay turnaround time by re-engineering Hamilton liquid handling codebase for lab-based workflows, directly improving experimental throughput.
- Established reliable multi-device communication by integrating laboratory instruments into robotic workflows via TCP/IP, with proactive monitoring to resolve connection failures before impacting runs.
- Authored python automation toolkit covering data transformation and analysis, eliminating repetitive manual tasks and standardizing outputs across R&D teams

Senior Research Associate | Dermibiont Inc.

Oct 2021 – Sep 2022

- Constructed end-to-end genomic data pipelines leveraging Nanopore technology and open-source bioinformatics tools to process high-throughput microbiome datasets in production.
- Partnered with bioinformatics engineers to refactor Dockerized QC data pipelines, achieving a 25% reduction in processing errors and raising data quality consistency across production sequencing runs.

PROJECTS

AI Hospital Bill Analyzer

- Building an AI-powered hospital bill analyzer using Gemini API and Tesseract OCR with Ollama to review medical invoices, designed to detect billing errors, flag overcharges, and generate itemized cost breakdowns for patients.

Personal Website

- Deployed a React portfolio website to showcase professional experience, technical projects, and programming.

EDUCATION

Northeastern University

Certificate in Bioinformatics, concentration in software development and data management

Boston, MA

May 2025

Uniwersytet Jagielloński

Master's in Molecular Biotechnology

Krakow, PL

Jul 2021